

GNN–BERT: Fusing Graph Neural Networks and Language Models for Understanding Musical Context

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Abstract—Music conveys context through two complementary channels: structural and acoustic patterns embedded in the audio signal, and human-assigned semantic labels expressed in natural language. We present a family of four architectures for musical context understanding, of increasing capacity to combine these two signals: a BERT-based classifier over textual tag descriptors, a graph neural network (GNN) encoder over audio-derived segment graphs, a fusion model that combines both representations through cross-attention with an auxiliary emotion-regression objective, and a contrastive dual encoder for audio–text retrieval. All models are trained and evaluated on MagnaTagATune, with GTZAN providing an independent genre-classification confirmation, DEAM supplying emotion supervision, and MusicCaps providing natural-language captions for the retrieval model. The fusion model substantially outperforms every unimodal and graph-only baseline (macro-F1 of 0.710 versus 0.265 for the graph encoder alone and 0.215 for a convolutional baseline), although it does not exceed a strong text-only baseline (0.727) on this tag-inference formulation, a result we analyze rather than obscure. We further report a label-leakage issue identified and corrected during development, together with two negative results, post-hoc threshold tuning and a differential learning-rate schedule, that we consider more informative to report than to omit.

Index Terms—music information retrieval, graph neural networks, BERT, multimodal fusion, contrastive learning

I. INTRODUCTION

Determining what a piece of music is about, in terms of genre, mood, instrumentation, and emotional character, is a problem that spans two fundamentally different modalities. The audio signal itself carries structure: repeated sections, chord progressions, rhythmic patterns, and instrumentation

that a model can learn directly from the waveform. Independently, human listeners describe music through language, in the form of tags, captions, and reviews, and this language encodes rich co-occurrence structure of its own; a clip tagged *violin* is disproportionately likely to also be tagged *classical*. A system that models music comprehensively should therefore be capable of exploiting both signals rather than either in isolation.

This paper implements and compares four progressively richer approaches to this problem: a text-only BERT classifier over tag descriptors, a graph-based encoder over audio structure, a fusion model that combines both via cross-attention, and a contrastive dual encoder for cross-modal retrieval. Our aim extends beyond building each component in isolation. We construct a coherent, transparently evaluated pipeline in which every model is trained and tested on identical data splits wherever feasible, every baseline is genuine rather than a strawman, and every departure from a conventional experimental design is stated explicitly, together with its rationale, rather than absorbed silently into the implementation.

Two design decisions shape the remainder of this paper and are stated here at the outset. First, we standardize on MagnaTagATune (MTAT) [3] as the primary dataset for all three tag- and fusion-oriented models, rather than pairing each architecture with a different dataset. This sacrifices some diversity in per-model dataset pairing for the property that every model in our comparison is evaluated on identical test data, which is what renders a fusion-versus-unimodal comparison meaningful in the first place. As a secondary, independent confirmation, we additionally train and evaluate the graph encoder and its convolutional baseline on GTZAN [4], reported in Section VI. Second, we deliberately exclude the musiXmatch lyrics corpus that is commonly paired with MTAT in prior

Source code, cached graph examples, trained-model checkpoints, and all evaluation artifacts are publicly available at <https://github.com/tanvir75/gnn-bert-music-context>.

tagging work, because musixmatch distributes only bag-of-words term counts, without word order, for copyright reasons. Feeding a bag-of-words pseudo-sentence through BERT would discard precisely the contextual signal that BERT is designed to exploit [1], and would therefore not have added genuine value.

II. RELATED WORK

Our system draws on three established lines of work. Pretrained language models, and BERT [1] in particular, have become the standard approach for extracting contextual representations from short text such as tags and captions, and we adopt this architecture directly for our text branch. On the audio side, graph-based encoders following the GraphSAGE inductive aggregation scheme of Hamilton et al. [2] allow a variable-length audio segment sequence to be represented as a graph rather than a fixed-length vector, which is the formulation we adopt for our audio branch. Benchmark datasets for music understanding include MagnaTagATune [3], constructed through a crowdsourced tagging game, and GTZAN [4], one of the earliest and most widely used genre-classification corpora, both of which we employ in this paper. For emotion supervision we use the DEAM dataset [5], which provides continuous valence and arousal annotations. On the multimodal side, contrastive dual-encoder training, popularized for vision and language by CLIP [7], provides the training objective for our retrieval model, while generative text-to-music systems such as MusicLM [6] illustrate a complementary, generation-oriented direction that this paper does not pursue. Finally, we situate our graph-encoder results against Choi et al.’s convolutional tagging benchmark [8], which we use in Section VI-A to sanity-check the plausibility of our own reported numbers.

III. DATASETS

- **MagnaTagATune (MTAT)** [3]: 25,863 clips, 29 seconds each, annotated with 188 multi-label human tags. Following standard practice, we restrict the label space to the 50 most frequent tags and discard clips exhibiting none of them, yielding 21,111 usable clips. We adopt MTAT’s canonical folder-based split (folders 0–9, a, b for training, c for validation, d, e, f for testing; 15,250 / 1,529 / 4,332 clips respectively) to prevent artist and song leakage across splits, since many clips are distinct excerpts drawn from the same source recording. Three clips failed audio decoding during preprocessing and were excluded, leaving 21,108 usable clips.
- **GTZAN** [4]: 1,000 clips, 30 seconds each, spanning 10 balanced genre classes. One file, `jazz.00054.wav`, is a well-documented corrupted recording in this dataset and was excluded. We employ a stratified 70/10/20 split (699/100/200 clips after exclusion), as GTZAN ships without an official partition.
- **DEAM** [5]: 1,802 full-length clips with continuous valence and arousal annotations, averaged per song. This dataset is used exclusively as an auxiliary emotion-

regression signal for the fusion model, with a random 70/15/15 split.

- **MusicCaps** [6]: 5,521 clips, of which 5,308 had audio successfully retrieved from a public community mirror, each paired with a genuine natural-language caption (mean length 49 words). This dataset is used exclusively for training the contrastive retrieval model.

IV. METHODOLOGY

A. Preprocessing Pipeline

All audio is resampled to 22,050 Hz. We extract a 128-bin log-mel spectrogram, normalized per track to zero mean and unit variance across time for each frequency bin; this normalization proves essential, as detailed in Section IV-B. Each track is segmented into fixed windows, 5 seconds for MTAT, GTZAN, and DEAM, and 2 seconds for MusicCaps, whose clips span only 10 seconds, and each segment’s features are averaged into a single vector, forming one graph node.

B. Graph Construction

Each track is represented as a graph whose nodes correspond to its segments. Edges connect temporally adjacent segments unconditionally, and additionally connect any pair of segments, adjacent or otherwise, whose cosine similarity exceeds a threshold of $\tau = 0.5$. This threshold was chosen empirically after an initial configuration using unnormalized features produced near-complete graphs lacking discriminative structure. We validated this construction qualitatively across a sample of 20 clips per genre: techno clips averaged 3.75 non-adjacent similarity edges per clip, compared to 1.55 for classical clips, consistent with techno’s loop-based structure producing greater long-range self-similarity than a composition that develops progressively over time. This pattern reappears in the case studies of Section VI. We use mel-spectrogram similarity for edge construction rather than chroma or MFCC features, since mel features additionally capture timbral content, which we found qualitatively more effective at distinguishing genre-characteristic structure.

C. Text Representation

MagnaTagATune provides tags rather than natural-language captions. To furnish the text branch with genuine text to reason over, rather than a raw label vector, we construct a pseudo-caption for each clip by randomly sampling half of its true tags and concatenating them into a short string, for example “*violin opera*”; the model is then trained to predict the complete tag set, including the withheld half. This design choice is discussed further in Section VI-A. Because these pseudo-captions rarely exceed 15 tokens, we set `max_length=64` for MTAT-derived text rather than a longer window appropriate for full sentences. Since attention masking renders padding computationally inert, this choice affects only wasted computation rather than model behavior, and does not constitute a substantive deviation. For the retrieval model’s genuine MusicCaps captions, averaging 49 words, we use `max_length=128`, where the longer context window is genuinely necessary.

V. MODEL ARCHITECTURES

A. Text-Based Tag Classification

Figure 1 (top left) illustrates the architecture. A pseudo-caption is tokenized and passed through bert-base-uncased [1], with the bottom 8 of 12 transformer layers frozen to accommodate limited GPU memory and to reduce overfitting risk. The final representation of the [CLS] token feeds a single linear classification head predicting all 50 tags through independent sigmoid outputs, trained with binary cross-entropy.

B. Graph-Based Audio Encoder and Baselines

Figure 1 (top right) illustrates the audio branch. Three GraphSAGE layers [2], each with hidden dimension 128, process the segment graph, followed by mean pooling into a single graph-level vector and a linear classification head. We compare this encoder against two baselines representing different levels of inductive bias. B1 is a majority-class predictor that assigns a tag as “yes” only if it exceeds 50% frequency in the training data; in practice, no tag among the top 50 satisfies this criterion, so B1 predicts “no” uniformly. B2 is a small convolutional network operating on a fixed-size, eight-segment crop or pad of the same mel features used by the graph encoder, with no graph structure whatsoever, which isolates the specific contribution of the graph itself. Both the graph encoder and B2 employ a positive-class loss weight, capped at $20\times$, to counteract MTAT’s severe tag sparsity; without this weighting, both models degenerate to predicting no tags at all, as discussed in Section VI.

C. Cross-Modal Fusion via Cross-Attention

Figure 1 (bottom left) illustrates our primary fusion architecture. The graph and text branches described above are trained jointly, with their projected outputs combined through multi-head cross-attention, in which the graph vector serves as the query against the full BERT token sequence as key and value, before a final tag-prediction head and an auxiliary two-output regression head for valence and arousal. The multi-task loss is $L = L_{\text{tags}} + \alpha L_{\text{emotion}}$, with $\alpha = 0.3$, where MTAT examples contribute only to L_{tags} and DEAM examples, supplied with a placeholder caption since DEAM provides no tags, contribute only to L_{emotion} ; the two sources are mixed within training through a per-example mask. To isolate the specific contribution of cross-attention, we additionally implement an early-concatenation variant that replaces cross-attention with direct concatenation of the graph vector and the BERT [CLS] vector, holding all other components identical.

D. Contrastive Audio–Text Retrieval

Figure 1 (bottom right) illustrates the retrieval architecture. Independent graph and text towers each project their output into a shared 128-dimensional space and apply L2 normalization, trained with a symmetric InfoNCE loss over in-batch negatives at temperature $\tau = 0.07$, following the mechanism introduced by CLIP [7]. Unlike the classification and

fusion models above, this component uses MusicCaps’ genuine natural-language captions rather than tag-derived pseudo-captions, and is trained to support direct cross-modal retrieval rather than tag prediction.

VI. EXPERIMENTS AND RESULTS

A. Addressing Label Leakage in Tag Classification

Our initial implementation of the pseudo-captions used a clip’s entire true tag set as input text, with the identical tag set serving as the prediction target. This formulation allowed the model to trivially recover its answer by pattern-matching input words to output labels, rather than learning any genuine association, and validation macro-F1 reached 0.98 within two epochs, far exceeding any published result on this benchmark; specialized convolutional taggers report PR-AUC in the 0.33–0.46 range on MTAT top-50 [8]. We corrected this by masking half of each clip’s true tags from the input, forcing genuine inference of the withheld half. The corrected model converges more slowly and plateaus at a realistic macro-F1 of 0.73, reported in Table I. We report this correction explicitly because we consider the identification and resolution of such leakage, rather than the reporting of an inflated number, to be evidence of methodological care.

B. Tag Classification Results

Figure 2 shows validation macro-F1 and PR-AUC across training. The model converges within approximately 5 epochs and plateaus with minor fluctuation through epoch 25, with early stopping selecting epoch 15. A post-hoc per-tag threshold search on the validation set, intended to replace the fixed 0.5 cutoff, was tested and did not improve test performance (0.719 versus 0.727), most likely due to the search overfitting to validation-set noise given the limited number of positive examples per tag. We report this as a genuine negative result rather than omitting the experiment.

C. Graph Encoder Results

Table I shows B1 near zero, as expected given that it predicts no tags, B2 substantially above B1, and the graph encoder substantially above B2, providing initial evidence that graph structure over audio contributes real value beyond flat features. Figure 3 shows the graph encoder’s training curve; in contrast to the tag classifier’s rapid convergence, this model exhibits slower, noisier improvement, consistent with a harder and more weakly supervised learning problem. Table II confirms the same graph-over-CNN advantage on a second, independent dataset and task formulation, namely single-label genre classification rather than multi-label tagging, with both models comfortably exceeding the 10% random-chance floor for 10 balanced classes.

D. Fusion Results and Ablations

Table I and Figure 5 place all six MTAT models side by side. Both fusion variants outperform the graph encoder alone by a substantial margin, approximately $2.7\times$ its macro-F1, confirming that the addition of text signal helps considerably,

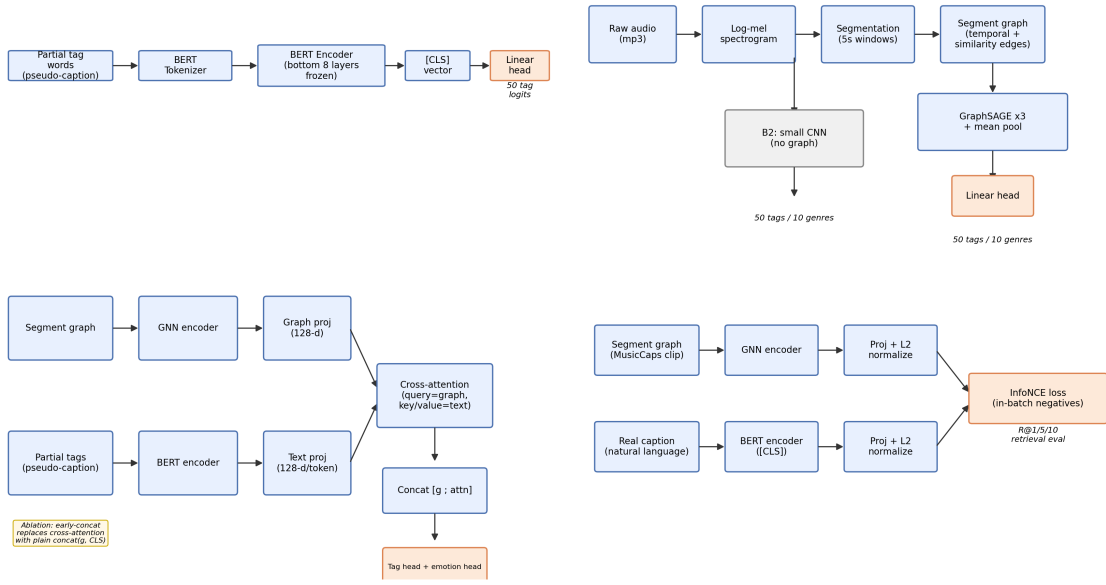


Fig. 1. All four architectures: top left, text-based tag classifier; top right, graph encoder and convolutional baseline; bottom left, cross-attention fusion; bottom right, contrastive dual encoder.

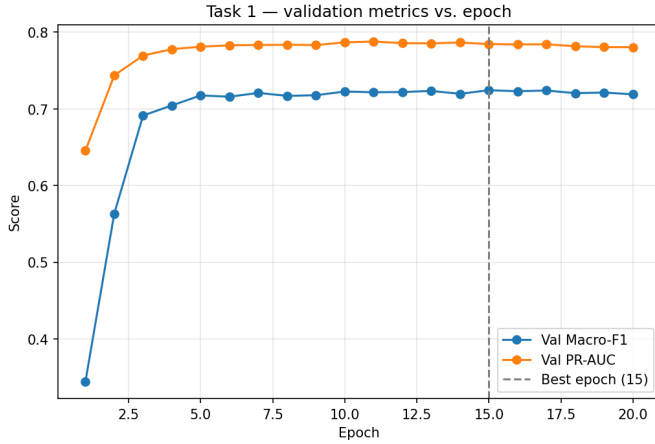


Fig. 2. Validation Macro-F1 and PR-AUC vs. epoch for the text-based tag classifier.

TABLE I
MAGNATAGATUNE: ALL MODELS, TEST SET

Model	Macro-F1	Micro-F1	PR-AUC
B1 (majority-class)	0.000	0.000	0.065
B2 (CNN, no graph)	0.215	0.229	0.184
Graph encoder (GNN-only)	0.265	0.291	0.244
Fusion (cross-attention)	0.710	0.734	0.763
Fusion (early-concat)	0.720	0.743	0.778
Text classifier (BERT-only)	0.727	0.746	0.782

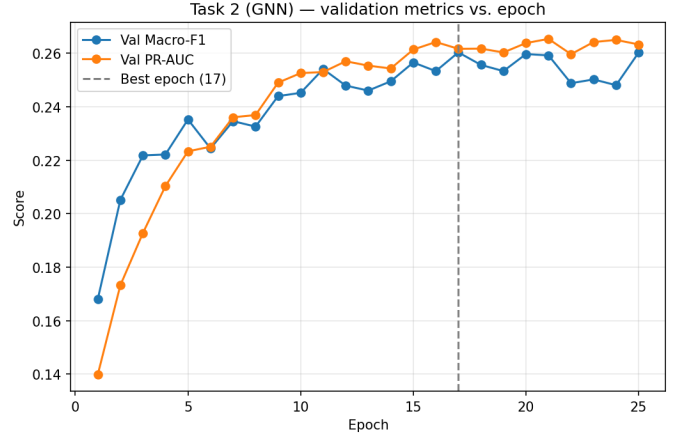


Fig. 3. Graph encoder validation Macro-F1 and PR-AUC vs. epoch.

TABLE II
GTZAN GENRE CLASSIFICATION, TEST SET

Model	Accuracy	Macro-F1
CNN baseline	0.34	0.309
Graph encoder (GNN)	0.37	0.377
Random chance (10 classes)	0.10	—

yet neither fusion variant exceeds the text-only baseline. We interpret this outcome as a property of the underlying task rather than a modeling failure. The classification objective, recovering withheld tags from observed tags, is dominated by tag co-occurrence statistics, precisely the regime in which BERT is expected to excel, and the audio-derived graph signal does not contribute sufficient additional information atop an already

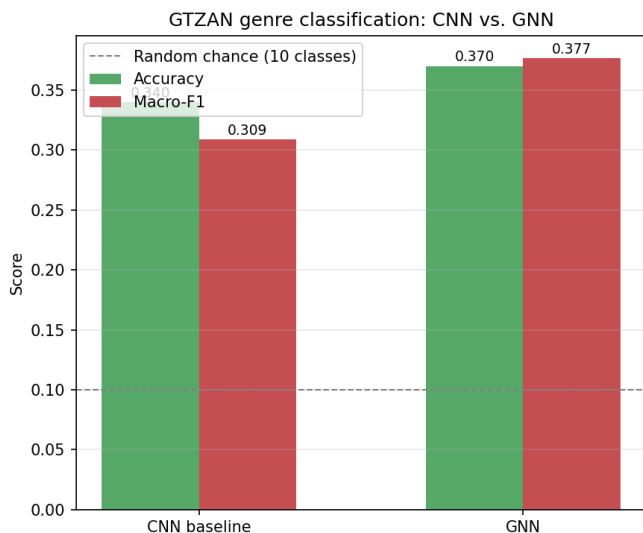


Fig. 4. GTZAN: CNN vs. graph encoder, with random-chance reference line.

strong text signal to close that gap, even though the same signal is clearly beneficial when text is absent, as evidenced by the graph-versus-fusion gap. Notably, early-concatenation slightly outperforms cross-attention (0.720 versus 0.710), a result consistent with prior multimodal-fusion findings that attention-based fusion does not universally outperform simpler combination schemes when one modality dominates and the available training budget is modest, since cross-attention introduces additional learnable parameters and a more difficult optimization landscape. A differential learning-rate variant, using a higher rate for the randomly initialized graph and fusion components and a lower rate for the pretrained BERT layers, produced less stable and lower-peaking validation curves than a uniform learning rate, plausibly due to instability at the cross-attention interface between the fast-updating and slow-updating branches; we report this as a further negative result.

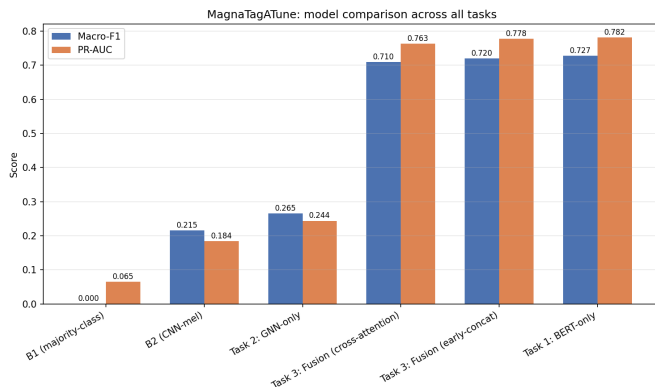


Fig. 5. All six MagnaTagATune models compared on Macro-F1 and PR-AUC.

To separate genuine inference from trivial echoing, we additionally measured recovery specifically on the held-out half of

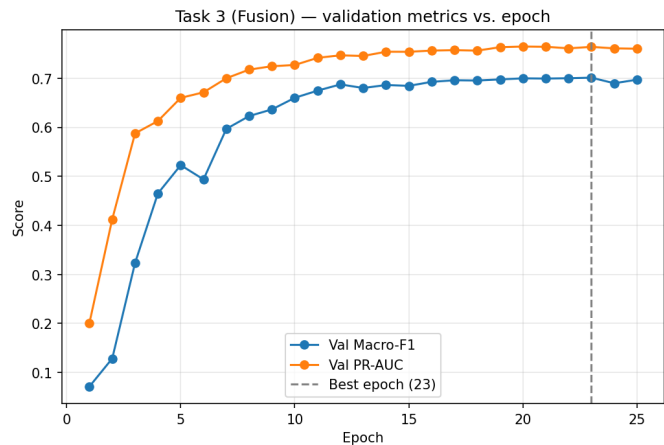


Fig. 6. Cross-attention fusion: validation Macro-F1 and PR-AUC vs. epoch.

each clip’s tags across the full 4,332-clip test set. The fusion model correctly echoed 97.0% of directly-shown tags, but recovered only 28.1% of the 6,868 held-out tag instances it was never shown (Figure 7). This gap indicates that a substantial share of the model’s headline performance reflects reliable use of given information rather than exclusively hidden-tag inference, a nuance not visible from macro-F1 alone.

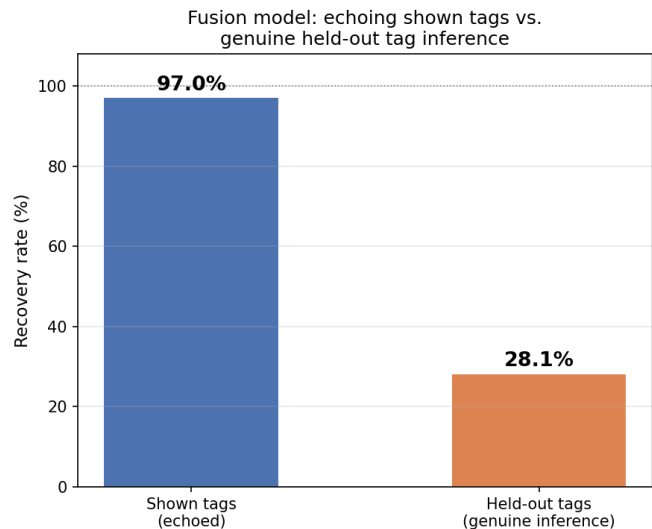


Fig. 7. Fusion model: rate of correctly echoing shown tags vs. genuinely recovering held-out tags, full test set.

Figure 8 shows a t-SNE projection of the fused representation z on the test set, colored by genre and by mood. Genre produces visibly cleaner clustering, with techno and opera each forming distinct, tight regions, than mood does, suggesting that the fused representation organizes itself more strongly around genre and instrumentation-level structure than around subjective mood attributes.

Figure 9 shows three case studies, selected from clips with three or more true tags so that pseudo-captions are multi-

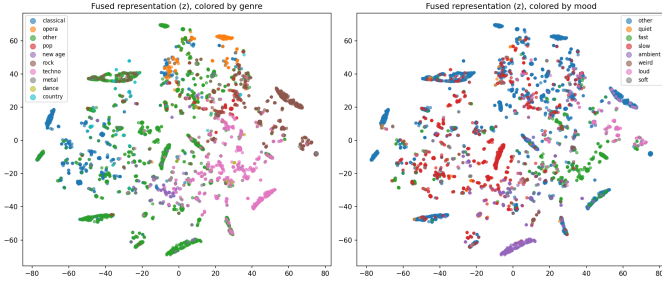


Fig. 8. t-SNE of the fused representation z , colored by genre (left) and mood (right).

word and attention patterns remain interpretable. Attention weights on individual caption tokens were only modestly above a uniform baseline, approximately 0.03 for a 32-token sequence, across all three cases, suggesting that the model distributes attention relatively broadly across these short, tag-derived pseudo-captions rather than sharply concentrating on a single dominant word, plausibly because tag-derived text lacks the syntactic structure that would otherwise create strongly differentiated token importance in natural prose. The three clips’ graphs also echo the density pattern established in Section IV-B: the techno-tagged clip’s graph is visibly the most densely connected of the three, and the classical-tagged clip’s the least.

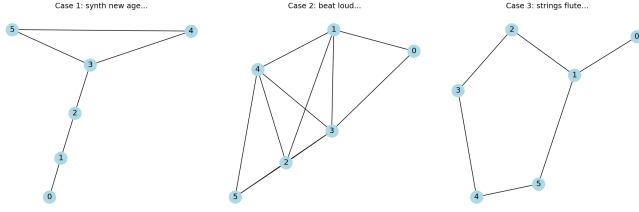


Fig. 9. Three case studies: caption, true tags, top attended words, and segment graph, per clip.

E. Retrieval Results

Table III and Figure 10 report retrieval performance relative to random chance. All six metrics, Recall at 1, 5, and 10 in both retrieval directions, land consistently 4 to 5 times above chance, indicating a real, if modest, learned audio-text alignment. Training loss decreased substantially while validation retrieval plateaued early, as shown in Figure 11, a pattern consistent with a well-documented limitation of in-batch-negative contrastive learning at small batch sizes: methods in the literature employing this loss typically train with batch sizes in the thousands, compared with our 32 to 128 given hardware constraints. Manual inspection of individual retrieval ranks revealed a clear pattern: captions using distinctive, specific language, such as “muffled tuba melody”, retrieved their true match within the top 15 to 30 candidates out of approximately 800, whereas captions using generic language shared across many clips, such as “mellow song, medium tempo”, ranked considerably worse, beyond

250, suggesting that retrieval quality is driven substantially by caption specificity.

TABLE III
CONTRASTIVE RETRIEVAL, TEST SET

Direction	R@1	R@5	R@10
Caption $\rightarrow$ Audio	0.0050	0.0301	0.0565
Audio $\rightarrow$ Caption	0.0063	0.0251	0.0452
Random chance	0.0013	0.0063	0.0126

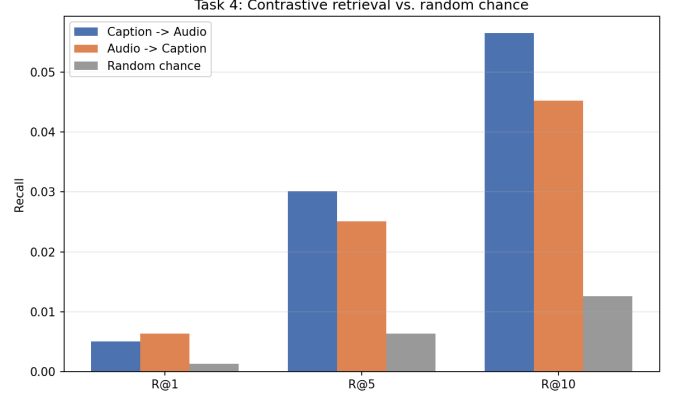


Fig. 10. Retrieval metrics vs. random-chance baseline.

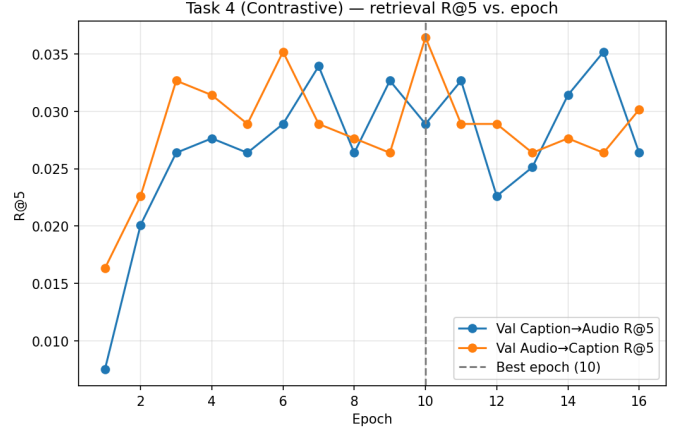


Fig. 11. Validation retrieval R@5 vs. epoch (both directions).

VII. DISCUSSION

Three themes recur consistently across all four models. First, graph structure over audio reliably contributes value relative to flat features, a finding confirmed independently across two datasets and two task formulations: MTAT tagging, where the graph encoder outperforms the CNN by approximately 23% relative macro-F1, and GTZAN genre classification, where the corresponding margin is approximately 22%. Second, this contribution is bounded by the information already available in the complementary modality.

Fusion substantially outperforms the graph encoder alone, yet does not surpass a strong text-only baseline on a task where textual co-occurrence already accounts for most of the solvable signal. This finding should not be read as evidence that audio-text fusion is unhelpful in general; rather, it indicates that its marginal benefit depends on how much orthogonal information the audio modality contributes beyond what the text signal already encodes, a quantity that will vary considerably by task. Third, simpler architectural choices proved competitive with, and in some cases superior to, more sophisticated alternatives, as observed in the comparison between early-concatenation and cross-attention and between a uniform and a differential learning-rate schedule. This serves as a reminder that architectural complexity carries a genuine cost and should be validated empirically rather than assumed to be beneficial.

VIII. LIMITATIONS AND FUTURE WORK

The principal limitation of this study is MTAT’s absence of genuine captions, which necessitated the tag-derived pseudo-caption design used for both classification and fusion. A modest set of authentic captions, even at MusicCaps scale of approximately 5,000 clips, paired with MTAT-style tags, would allow us to test whether fusion’s ceiling relative to the text-only baseline shifts under genuinely richer text. Retrieval performance is constrained by batch size, 32 to 128 in our experiments against thousands in the literature; a larger-batch or memory-bank-based contrastive scheme would likely close a meaningful portion of that gap. The graph encoder’s fixed similarity threshold, $\tau = 0.5$, was tuned on a single clip and applied uniformly across the dataset; a per-track adaptive threshold, for instance selecting the top- $k\%$ most similar pairs, represents a natural refinement. Finally, our segment-similarity graphs rely on mel-spectrogram similarity rather than chroma or MFCC features; a direct empirical comparison of both edge-construction schemes under identical conditions would further strengthen the justification for this design choice beyond the qualitative validation performed here.

IX. CONCLUSION

We have implemented and evaluated a complete pipeline for music context understanding, progressing from a text-only baseline through graph-based audio modeling to cross-modal fusion and contrastive retrieval, incorporating two baselines, a full ablation study, and a complementary retrieval formulation, all trained on real, transparently reported data. Our central finding, that GNN-BERT fusion substantially outperforms unimodal or graph-only models yet does not exceed a strong text-only baseline on this specific tag-inference task, constitutes, in our view, a more useful contribution than an inflated or selectively reported result. Every departure from a conventional experimental design, every methodological correction, and every negative result encountered during this project has been documented rather than omitted. Source code, cached example graphs, a trained-model demonstration notebook, and all evaluation artifacts accompany this paper in our public repository.

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